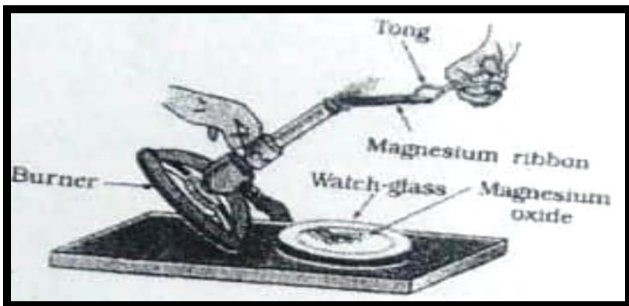


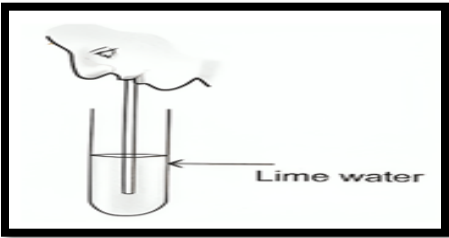
MAX MARKS: 80 TIME ALLOWED: 3 HRS

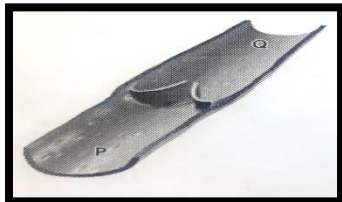
General Instructions:**Read the following instructions carefully and strictly follow them:**

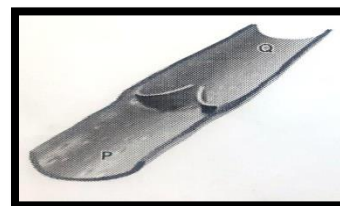
- (i) This question paper comprises 39 questions. All questions are compulsory.
- (ii) This question paper is divided into five sections - A, B, C, D and E.
- (iii) Section A - Questions No. 1 to 20 are Multiple Choice questions. Each question carries 1 mark.
- (iv) Section B - Questions No. 21 to 26 are Very Short Answer type questions. Each question carries 2 marks.
- (v) Section C - Questions No. 27 to 33 are Short Answer type questions. Each question carries 3 marks.
- (vi) Section D - Questions No. 34 to 36 are Long Answer type questions. Each question carries 5 marks.
- (vii) Section E - Questions No. 37 to 39 are Case based units of assessment carrying 4 marks each with sub-parts.

SECTION - A

SECTION - A		
1	Which of the following is correct observation of the reaction shown in above set up? (a) Brown powder of magnesium oxide is formed. (b) Colourless gas which turns lime water milky is evolved. (c) Magnesium ribbon burns with brilliant white light. (d) Reddish brown gas with a smell of burning sulphur has evolved.	1
		
2	In the redox reaction, choose the correct statement. $\text{MnO}_2 + 4\text{HCl} \rightarrow \text{MnCl}_2 + 2\text{H}_2\text{O} + \text{Cl}_2$ (a) MnO_2 is reduced to MnCl_2 & HCl is oxidized to H_2O. (b) MnO_2 is reduced to MnCl_2 & HCl is oxidized to Cl_2. (c) MnO_2 is oxidized to MnCl_2 & HCl is reduced to Cl_2. (d) MnO_2 is oxidized to MnCl_2 & HCl is reduced to H_2O.	1
3	When a small amount of acid is added to water, the phenomenon that occurs is: (i) Dilution (ii) Neutralization (iii) Formation of H_3O^+ ions (iv) Salt formation The correct statements are : (a) (i) and (iii) (b) (ii) and (iv)	1

	(c)(i) and (ii) (d)(iii) and (iv)	
4	Identify the basic salt from the following salts: (a)Na ₂ CO ₃ (b)NH ₄ Cl (c)NaNO ₃ (d)KCl	1
5	Which one of the following properties is not generally exhibited by ionic compounds? (a)Solubility in water (b)Electrical conductivity in solid state (c)High melting and boiling points (d)Electrical conductivity in molten state	1
6	Pentane has the molecular formula C ₅ H ₁₂ .It has (a)5 covalent bonds (b)12 covalent bonds (c)16 covalent bonds (d)17 covalent bonds	1
7	Generally food is broken and absorbed within the body of organisms. In which of the following organisms is it done outside the body? (a)Amoeba (b)Mushroom (c)Paramecium (d)Lice	1
8	Observe the activity given below. What does it help to conclude, when the person exhales into the test-tube? (a)Percentage of carbon dioxide is more in inhaled air (b)Fermentation occurs in the presence of oxygen. (c)Percentage of carbon dioxide is more in the exhaled air. (d)Fermentation occurs in the presence of carbon dioxide.	1
		
9	Sensory nerve of a reflex arc carries information from the receptor cells to the: (a)spinal cord (b)brain (c)muscles of the effector organ (d)bones of the receptor organ	1
10	A student while observing an embryo of a gram seed listed various parts of the embryo as listed below: Testa, Micropyle, Cotyledon, Tegmen, Plumule, and Radicle. On examining the list, the teacher commented that only three parts are correct. Select these three correct parts: (a)Cotyledon,Testa,Plumule (b)Cotyledon,Plumule,radicle (c)Cotyledon,Tegmen,radicle (d)Cotyledon,Micropyle,Plumule	1
11	An attached earlobe in humans is an inherited condition. The allele for attached earlobes is, recessive.What are the chances of parents, both having attached earlobes, to have a child with attached earlobes? (a)0% (b)25% (c)75% (d)100%	1
12	Humans have two different sex chromosomes,X and Y.Based on Mendel's laws; a male offspring will inherit which combination of chromosomes? (a)Both the X chromosomes from one of its parents. (b)Both the Y chromosomes from one of its parents. (c)Combination of X chromosomes from either of its parents. (d)Combination of x and Y chromosome from either of its parents.	1
13	Two bulbs of 100W and 40W are connected in series. The current through the 100W bulb is 1A.The current through the 40 W bulb will be: (a)0.4A (b)0.6A (c)0.8A (d)1A	1
14	A cylindrical conductor of length 'l' and uniform area of cross section 'A' has resistance 'R'.The area of cross section of another conductor of same material and same resistance but of length '2l' is: (a)A/2 (b)3A/2 (c) 2A (d)3A	1

15	Which of the following pattern correctly describes the magnetic field around a long straight wire carrying current? (a) Straight lines perpendicular to the wire.(b) Straight lines parallel to the wire. (c) Radial lines originating from the wire.(d) Concentric circles centered around the wire.	1										
16	Suppose a horseshoe magnet is held vertically upwards with the North Pole is on left. A wire passing between the poles carries a direct current directed away from you. In what direction is the magnetic force exert on the wire? (a)Downward (b)Upward (c)Towards east (d)Towards South	1										
	For questions number 17 to 20,two statements are given - one labelled as Assertion (A) and the other labelled as Reason (R).Select the correct answer to these questions from the codes (A), (B), (C) and (D) as given below. (A)Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion(A) (B)Both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of Assertion(A) (C)Assertion (A) is true, but Reason (R) is false. (D)Assertion (A) is false, but Reason (R) is true.											
17	Assertion (A): Silver bromide decomposition is used in black and white photography. Reason(R): Light provides energy for this exothermic reaction.	1										
18	Assertion (A): After white washing the walls, a shiny white finish on walls is obtained after two to three days. Reason(R): Calcium oxide reacts with carbon dioxide to form calcium hydrogen carbonate, which gives a shiny white wash.	1										
19	Assertion (A): Height in pea plants is controlled by efficiency of enzymes and is thus genetically controlled. Reason(R): Cellular DNA is the information source for making proteins in the cell.	1										
20	Assertion (A): When a solenoid carries a clockwise current, one end acts as the south pole, while the other end acts as the north pole. Reason(R): The magnetic field inside a solenoid runs from the south pole to the north pole.	1										
	SECTION - B											
21	Keerti added dilute hydrochloric acid to four metals and recorded her observations as shown in the table given below: <table><tr><th>Metal</th><th>Gas evolved</th></tr><tr><td>Copper</td><td>Yes</td></tr><tr><td>Iron</td><td>Yes</td></tr><tr><td>Magnesium</td><td>No</td></tr><tr><td>Zinc</td><td>yes</td></tr></table> Select the correct observation(s) and give chemical equation(s) of the reaction involved.	Metal	Gas evolved	Copper	Yes	Iron	Yes	Magnesium	No	Zinc	yes	2
Metal	Gas evolved											
Copper	Yes											
Iron	Yes											
Magnesium	No											
Zinc	yes											
22	State the events occurring during the process of photosynthesis. Is it essential that these steps take place one after the other immediately?	2										
23	The image below shows the cross section of a blood vessel of a human arm. (a)What is the type of blood vessel shown in the image? (b)Which direction will the blood flow in such a blood vessel? 	2										



24	Select anyone among A and B A] Different parts of brain are associated with specific functions. Name the part of human brain which performs the following functions: (a) Sensation of feeling full (b) Vomiting (c) Picking up a pencil (d) Riding a bicycle OR B] List two differences between the movement of leaves of a sensitive plant and the movement of a shoot towards light.	2
25	Select anyone among A and B A] Give reasons for the following: (a) Red traffic signals can be seen from a very long distance. (b) Stars appear to be slightly higher than their actual position. OR B] A student sitting at the back bench in a class, is not able to see what is written on the blackboard. He, however, sees it clearly when sitting on the front seat at an approximate distance of 1.5 cm from the blackboard. Draw ray diagrams to illustrate the image formation of the blackboard when he is seated at the (a) back seat (b) front seat.	2
26	In the following food chain, only 2 J of energy was available to the peacocks. How much energy would have been present in Grass? Justify your answer. Grass----->Grasshopper----->Frog----->Snake----->Peacock	2
SECTION - C		
27	When solutions of silver nitrate and sodium chloride are mixed, a white precipitate forms. The ionic equation for the reaction is $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s})$ (i) (a) What is the name of the white precipitate? (b) Is it a soluble or insoluble compound? (ii) Is the precipitation of silver chloride a redox reaction?	3
28	Select anyone among A and B A] (a) Write electron dot structures of Ca (atomic no.20) and O (atomic no.8). (b) Show the formation of calcium oxide by transfer of electrons. (c) Name the ions present in this compound. OR B] A teacher asks her students to identify a metal; M. She gives them the following clues to help them. (P) Its oxide reacts with both HCl and NaOH (Q) It does not react with hot or cold water but reacts with steam. (R) It can be extracted by electrolysis of its ore. (a) Identify the metal. (b) Write the chemical equations for the reaction of the metal with HCl and NaOH respectively. (c) What would happen if the metal is reacted with iron oxide?	3
29	Complete the following flowchart as per the given instructions. <pre> graph TD A[Gastric Glands Present in the wall of Stomach] -- Secretions --> B[a] A -- Secretions --> C[b] A -- Secretions --> D[c] B -- Function --> E[d] C -- Function --> F[e] D -- Function --> G[f] </pre>	3
30	(a) Water has refractive index 1.33 and alcohol has refractive index 1.36. Which of the two medium is optically denser? Give reason for your answer. (b) Draw a ray diagram to show the path of a ray of light passing obliquely from water to alcohol.	3

31	A student holding a mirror in his hand directed the reflecting surface of the mirror towards the Sun. He then directed the reflected light on to a sheet of paper held close to the mirror. (a) What should he do to burn the paper? (b) Which type of mirror does he have? (c) Will he be able to determine the approximate value of focal length of this mirror from this activity? Give reason and draw ray diagram to justify your answer in this case.	3
32	A compass needle is placed near a current carrying wire. State your observations for the following cases and give reasons for the same in each case: (a) Magnitude of electric current in wire is increased. (b) The compass needle is displaced away from the wire.	3
33	Gas A, found in the upper layers of the atmosphere, is a deadly poison but is essential for all living beings. The amount of this gas started declining sharply in the 1980s. (a) Identify gas A. How is it formed at higher levels of the atmosphere? (b) Why is it essential for all living beings? State the cause for the depletion of this gas.	3
SECTION - D		
34	<u>Attempt either option A or B</u> A] An organic compound P is a constituent of wines. P on reacting with acidified $K_2Cr_2O_7$ forms another compound Q. When a piece of sodium is added to Q, a gas R evolves which burns with a pop sound when a burning match stick is brought near it. (a) Give the chemical name of compound P. (b) Mention another use of compound P apart from the use mentioned in the question. (c) Illustrate with the help of chemical equation the conversion of P into Q. (d) Give a balanced equation to depict the reaction of Q with sodium. (e) What happens when P is heated with conc. H_2SO_4 at 443K? Write its chemical equation. OR B] An organic compound X is liquid at room temperature. It is also a very good solvent and has the molecular formula C_2H_6O. Upon oxidation X gives Y. Y releases a gas W with brisk effervescence on reacting with $NaHCO_3$. X reacts with Y in the presence of conc. H_2SO_4 to give another compound Z which has a pleasant smell. (a) Give the chemical name and chemical formula of Y. (b) How will you test for the gas W? (c) Depict the formation of Y and Z using chemical equations. (d) Name the reaction of formation of Z. (e) Give any one use of Z.	5
35	<u>Attempt either option A or B</u> A] (a) Draw a diagram to show spore formation in Rhizopus. (b) With the help of an example differentiate between the process of budding and fragmentation. (c) Why is vegetative propagation practiced for growing some type of plants? OR B] (a) Draw the diagram of female reproductive system and match and mark the part(s): (i) Where block is created surgically to prevent fertilization. (ii) Where CuT is inserted? (iii) Inside which condom can be placed? (b) Why do more and more people prefer to use condoms? What is the principle behind use of condoms?	5
36	<u>Attempt either option A or B</u> A] (a) What is meant by the statement. "The resistance of a conductor is one ohm"? (b) Define electric power. Write an expression relating electric power, potential difference and resistance. (c) How many 132 ohm resistors in parallel are required to carry 5A on a 220 V line? OR B] (a) An electric iron consumes energy at a rate of 880 W when heating is at the maximum rate and	5

	330 W when heating is at the minimum. If the source voltage is 220 V, calculate the current and resistance in each case. (b)What is the heating effect of electric current? (c)Find an expression for the amount of heat produced when a current passes through a resistor for some time.	
	SECTION - E	
37	On the basis of reactivity, metals are grouped into three categories: (a) Metals of low reactivity (b) Metals of medium reactivity (c) Metals of high reactivity Therefore, metals are extracted in pure form from their ores on the basis of their chemical properties. Metals of high reactivity are extracted from their ores by electrolysis of the molten ore. Metals of low reactivity are extracted from their sulphide ores, which are converted into their oxides. The oxides of these metals are reduced to metals by simple heating. (a)Name the process of reduction used for a metal that gives a vigorous reaction with air and water both. (b)Carbon cannot be used as a reducing agent to obtain aluminium from its oxide? Why? <u>Attempt either option c or d</u> (c)Describe briefly the method to obtain mercury from Cinnabar. Write the chemical equation for the reactions involved in the process. OR (d)Differentiate between roasting and calcinations giving a chemical equation for each.	1 1 2
38	<u>Attempt either option A or B</u> (A)The most obvious outcome of the reproductive process is the generation of individuals of similar design, but in sexual reproduction they may not be exactly alike. The resemblances as well as differences are marked. The rules of heredity determine the process by which traits and characteristics are reliably inherited. Many experiments have been done to study the rules of inheritance. (a)Why an offspring of a human being is not a true copy of his parents in sexual reproduction? (b) While performing experiments on inheritance in plants, what is the difference between F₁ and F₂ generation? <u>Attempt either option c or d</u> (c)Why do we say that variations are useful for the survival of a species over time? OR (d)Study Mendel's cross between two plants with a pair of contrasting characters. RRYY X rryy Round Yellow Wrinkled Green He observed 4 types of combinations in F₂ generation. Which of these were new combinations? Why do new features which are not present in the parents, appear in F₂ generation? OR (B)Sex of an individual is determined by different factors in various species. Some animals rely entirely on the environmental cues, while in some other animals the individuals can change their sex during their life time indicating that sex of some species is not genetically determined. However in human beings, the sex of an individual is largely determined genetically. (a)In what way the sex chromosomes X and Y different in size? Name the mismatched pair of sex chromosome in humans. (b)Write the number of pair/pairs of sex chromosomes present in human beings. In which one of the parent (male/female) perfect pair/pairs of sex chromosome are present? (c)Citing two examples, justify the statement"Sex of an individual is not always determined genetically". OR (c)Draw a flow chart to show that sex is determined genetically in human beings.	1 1 2 2

Attempt either option A or B

A] A student took three concave mirrors of different focal lengths and formed the experiment to see the image formation by placing an object in different distances with these mirrors as shown in the following table:

Case no.	Object distance	Focal length
I	45 cm	20 cm
II	30 cm	15 cm
III	20 cm	30 cm

(a) List two properties of the image formed in Case I.

(b) In which one of the cases given in the table the mirror will form real image of same size and why?

(c) Name the type of mirror used by dentists. Give reason on why do they use such type of mirrors.

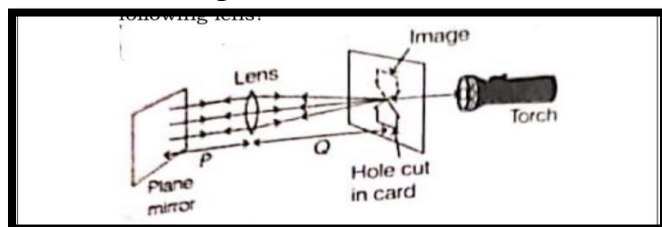
OR

(c) Look at the table and identify the situation (object distance and focal length) which resembles the situation in which concave mirrors are used as shaving mirrors? Draw a ray diagram to show the image formation in this case.

OR

B] Magnification products like magnifying glass, compound microscope, telescope, etc., are important instruments for the daily activities of many individuals today. Whether the application is for commercial, professional or personal use, a magnifier enhances one's ability to perform or enjoy a task or hobby. A magnifying glass is a convex lens that is used to produce a magnified image of an object. The magnification of a magnifying glass depends upon whether it is placed between the user's eye and the object being viewed and the total distance between them. The highest magnifying power is obtained by putting the lens very close to the eye and moving the eye and the lens together to obtain the best focus.

(a) The following diagram shows an experiment to measure the focal length of a lens. What is the focal length of the following lens?



(b) On what factors, magnification of a magnifying glass depends?

(c) Draw a ray diagram showing that the convex lens forms a virtual, erect, enlarged image of an object.

OR

(c) A convex lens forms an image of magnification-2 of height 6 cm. Find the height of the object.
